

VD3D: TAMING LARGE VIDEO DIFFUSION TRANSFORMERS FOR 3D CAMERA CONTROL

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<https://snap-research.github.io/vd3d>



Figure 1: **3D camera control for text-to-video generation.** We introduce a method that can control camera poses for text-to-video generation using video diffusion transformers. (left) The method takes as input a set of camera poses used to generate each frame of a rendered video. (center, right) Applying multiple camera trajectories with the same text prompt enables synthesis of complex scenes from a varied set of viewpoints. Video results: <https://snap-research.github.io/vd3d>.

ABSTRACT

Modern text-to-video synthesis models demonstrate coherent, photorealistic generation of complex videos from a text description. However, most existing models lack fine-grained control over camera movement, which is critical for downstream applications related to content creation, visual effects, and 3D vision. Recently, new methods demonstrate the ability to generate videos with controllable camera poses—these techniques leverage pre-trained U-Net-based diffusion models that explicitly disentangle spatial and temporal generation. Still, no existing approach enables camera control for new, transformer-based video diffusion models that process spatial and temporal information jointly. Here, we propose to tame video transformers for 3D camera control using a ControlNet-like conditioning mechanism that incorporates spatiotemporal camera embeddings based on Plücker coordinates. The approach demonstrates state-of-the-art performance for controllable video generation after fine-tuning on the RealEstate10K dataset. To the best of our knowledge, our work is the first to enable camera control for transformer-based video diffusion models.

1 INTRODUCTION

Text-to-video foundation models achieve unprecedented visual quality (Brooks et al., 2024; Sharma et al., 2024). They are trained on massive collections of images and videos and learn to synthesize remarkably consistent and physically plausible visualizations of the world. Yet, they lack built-in mechanisms for explicit 3D control during the synthesis process, requiring users to manipulate outputs through prompt engineering and trial and error—a slow, laborious, and computationally expensive process. For example, as Fig. 2 shows, state-of-the-art video models struggle to follow even simple “zoom-in” or “zoom-out” camera trajectories using text prompt instructions (see supplemental webpage). This lack of controllability limits interactivity and makes existing video generation

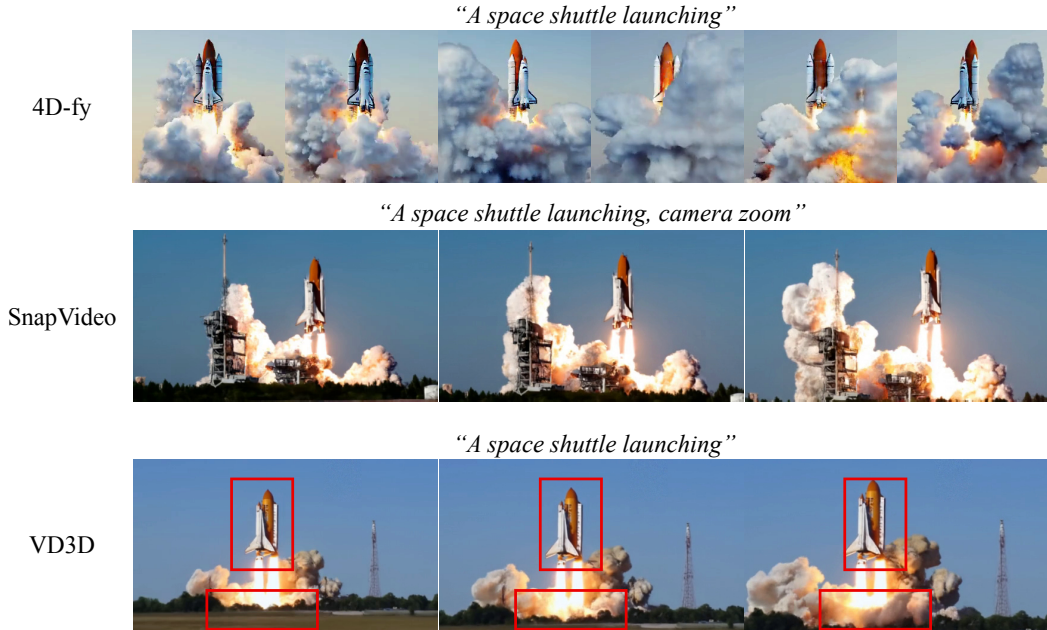


Figure 2: **Comparing text-to-video, text-to-4D, and camera-conditioned text-to-video generation.** We show time progressing across columns for generated videos from different approaches. Text-to-4D approaches, such as 4D-fy (Bahmani et al., 2024b) (top), have complete control over the camera through a 3D representation but lack photorealism. We visualize time progressing while simultaneously changing the viewpoint. (middle) Methods for text-to-video generation (Menapace et al., 2024) create realistic videos but do not provide explicit control over the viewpoint. In contrast, camera-conditioned text-to-video generation (bottom) bridges the gap between the two paradigms by extending text-to-video generators with 3D camera control without using an explicit 3D representation. Specifically, we condition VD3D with a zoom-in camera trajectory to control the generation.

techniques challenging to use for artists or other end users. We augment 2D video generation models with control over the position and orientation of the camera, providing finer-grained control compared to text prompting, and facilitating use of video generation models for downstream applications.

Several contemporary works (Wang et al., 2023e; He et al., 2024a; Yang et al., 2024b) propose methods for camera control of state-of-the-art, open-source video diffusion models. The key technical insight proposed by these methods is to add camera control by fine-tuning the temporal conditioning layers of a U-Net-based video generation model on a dataset with high-quality camera annotations. While these techniques achieve promising results, they are not applicable to more recent, high-quality transformer-based architectures (Vaswani et al., 2017; Peebles & Xie, 2023), such as Sora (Brooks et al., 2024), SnapVideo (Menapace et al., 2024), and Lumina-T2X (Gao et al., 2024a), as these latest works simply do not have standalone temporal layers amenable to camera conditioning.

Large video transformers represent a video as a (possibly compressed) sequence of tokens, applying self-attention layers to all the tokens jointly (Brooks et al., 2024; Menapace et al., 2024). Consequently, as diffusion transformers do not have standalone temporal layers, they are incompatible with current camera conditioning approaches. As the community shifts towards large video transformers to jointly model spatiotemporal dependencies in the data, it is critical to develop methods that provide similar capabilities for camera control. Our work designs a camera conditioning method tailored to the joint spatiotemporal computation used in large video transformers and takes a step towards taming them for controllable video synthesis.

We develop our work on top of our implementation of SnapVideo (Menapace et al., 2024), a state-of-the-art video diffusion model, that employs Far-reaching Interleaved Transformers (FIT) blocks (Chen & Li, 2023) for efficient video modeling in the compressed latent space. We investigate various camera conditioning mechanisms in the fine-tuning scenario and explore trade-offs in terms of visual quality preservation and controllability. Our findings reveal that simply adapting existing approaches

to video transformers does not yield satisfactory results: they either enable some limited amount of control while reducing the visual quality of the output video, or they entirely fail to control camera motion. Our key technical insight is to enable the control through spatiotemporal camera embeddings, which we derive by combining Plücker coordinates with the network input through a separately trained cross-attention layer. To the best of our knowledge, our work is the first to explore a ControlNet-like (Zhang et al., 2023) conditioning mechanism for spatiotemporal transformers.

We evaluate the method on a collection of manually crafted text prompts and unseen camera trajectories and compare to baseline approaches that incorporate previous camera control methods into a video transformer. Our approach achieves state-of-the-art results in terms of camera controllability and video quality, and also enables downstream applications such as multi-view, text-to-video generation, as depicted in Figure 1. In contrast to existing image-to-3D methods (e.g., (Liu et al., 2024a; Qian et al., 2024b; Voleti et al., 2024a)), which are limited to object-centric scenes, our approach synthesizes novel views for real input images with complex environments.

Overall, our work makes the following contributions.

- We propose a new method to tame large video transformers for 3D camera control. Our approach uses a ControlNet-like conditioning mechanism that incorporates spatiotemporal camera embeddings based on Plücker coordinates.
- We thoroughly evaluate this approach, including comparisons to previous camera control methods, which we adapt to the video transformer architecture.
- We show state-of-the-art results in camera-controllable video synthesis by applying the proposed conditioning method and fine-tuning scheme to the SnapVideo-based model (Menapace et al., 2024).

2 RELATED WORK

Our method is connected to techniques related to text-to-video, text-to-3D, and text-to-4D generation. As this is a popular and fast-moving field, this section provides only a partial overview with a focus on the most relevant techniques; we refer readers to Po et al. (2023) and Yunus et al. (2024) for a more thorough review of related techniques.

Text-to-video generation. Our work builds on recent developments in 2D video generation models. One such class of these techniques works by augmenting text-to-image models with layers that operate on the temporal dimension to facilitate video generation (Blattmann et al., 2023b; Singer et al., 2023a; Wu et al., 2023; Guo et al., 2024; Blattmann et al., 2023a). Video generation models can be trained in a hybrid fashion on both images and videos to improve the generation quality (Bain et al., 2021; Wang et al., 2023b; Xue et al., 2022; Ho et al., 2022a; Guo et al., 2024; He et al., 2022; Wang et al., 2023c; Zhou et al., 2022). While they are primarily based on convolutional, U-Net-style architectures, a recent shift towards transformer-based architectures enables synthesis of much longer videos with significantly higher quality (Brooks et al., 2024; Ma et al., 2024b; Menapace et al., 2024; Ma et al., 2024a). Still, these methods do not enable synthesis with controllable camera motion.

4D generation. Previous methods also tackle the problem of 4D generation, i.e., generating videos of dynamic 3D scenes from controllable viewpoints, usually from an input text prompt or image. Since the initial work on this topic using large-scale generative models (Singer et al., 2023b), significant improvements in the visual quality and motion quality of generated scenes have been achieved (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024a; Bahmani et al., 2024a). While these methods generate scenes based on text conditioning, other approaches convert an input image or video to a dynamic 3D scene (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024a; Ling et al., 2024a; Gao et al., 2024b; Zeng et al., 2024; Chu et al., 2024). Another line of work (Bahmani et al., 2023a; Xu et al., 2023) extends 3D GANs into 4D by training on 2D videos, however the quality is limited and models are trained on single category datasets. All of these methods are focused on object-centric generation, typically based on 3D volumetric representations. As such, they typically do not incorporate background elements, and overall, they do not approach the level of photorealism demonstrated by the state-of-the-art video generation models used in our technique (see Fig. 2).